Disorders of the Avian Gastrointestinal Tract

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Avian & Exotic Animal Medicine and Surgery

UCDAVIS VETERINARY MEDICINE

EXCEPTIONAL CARE, EDUCATION *and* INNOVATION

Learning Objectives

- Identify the most common G I disorders of birds and reptiles
- Order and interpret most appropriate diagnostic test(s) for common G I disorders
- Develop targeted treatment plans for common G.I. disorders based on analysis of clinical data

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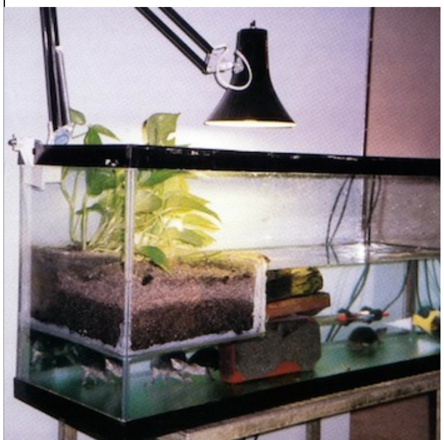
Image from thefunnycats.com

Patient Presentation



- Signalment
- Presenting complaint
- History
  - environment
  - diet
  - toxins
  - foreign body
  - images of cage

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History Questions should include (at minimum):

- Duration/ Progression of signs
- Appetite, fecal changes, regurgitation?
- Diet: include brand, fresh foods, table foods, supplements, water source
- Husbandry: Access to foreign material, Cage type/age, toy types, Exposure to other birds, outside, Risk of infectious disease (bird shows, boarding?)
- Medical history, previous medications

Baby birds not yet weaned; whether parent or hand reared

Excrement Evaluation IMPORTANT!

- 3 components
- Color
- Volume
- Consistency
- Undigested food
- "Stress" polyuria in birds

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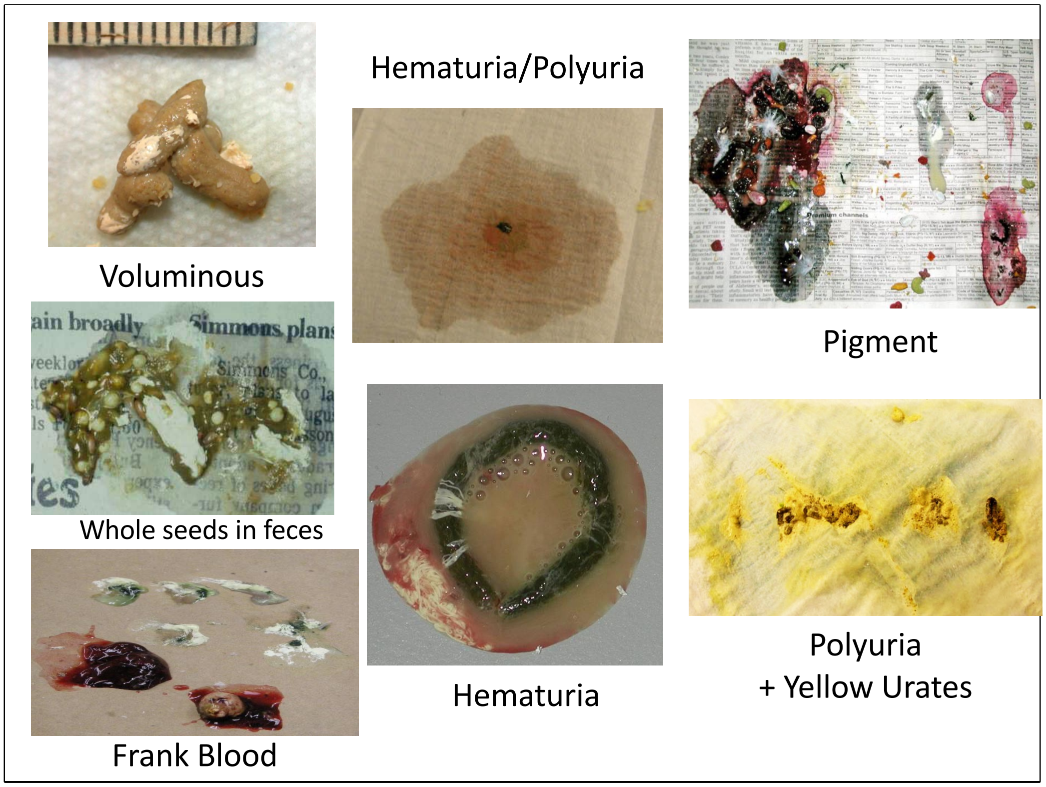
Bird and terrestrial herptiles droppings consist of three components: feces, urates and urine. The fecal component should be dark brown or slightly green (depending on diet), this may change with some food items (berries, beets, colored pellets).

Urates should be white and should be less volume than the fecal component.

- The urine is normally clear and only a small amount of liquid urine is passed with the droppings,
- **However, stress polyuria occurs when birds are put into stressful situations, like taken for car rides and taken to the vet!**
- This normally occurs due to the lack of normal retroperistalsis of urine into the coprodeum for fluid resorption, not through increased fluid production. Be sure to distinguish this from diarrhea!

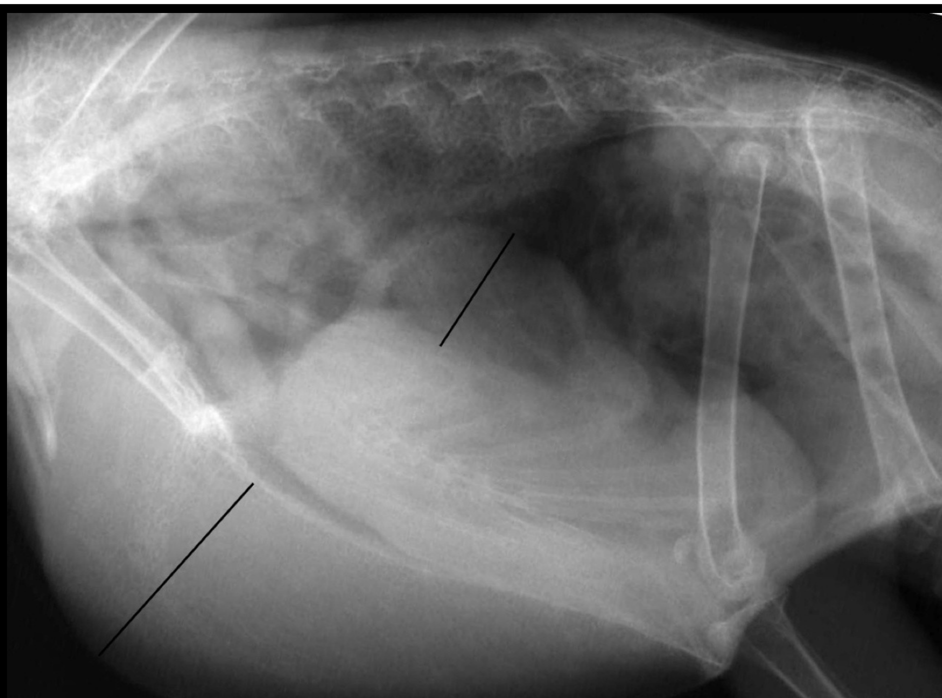
Larger droppings may be seen first thing in the morning for psittacines and in laying hens due to fecal retention.



- Evaluation of the feces, urates and urine is a integral part of every avian/reptile physical examination!
- Voluminous stools may indicate maldigestion or malabsorption. This may result from intestinal or pancreatic disease.
- Hematuria can be difficult to differentiate from hematochezia or melena. It is important to remember that excreted blood can come from the gastrointestinal, urinary or genital systems and all should be investigated. Frank blood clots in the droppings often indicate cloacal disease, but disease from the GI, reproductive or urinary tracts should be considered.
- Dark stools indicative of melena should give cause for suspicion of upper gastrointestinal hemorrhage. Pigmented feces can be difficult to differentiate from a pathologic process. It is important to find out if the bird was fed berries, beets or eats a particular colored pellet. This will quickly resolve if the suspected food item is removed. Stress polyuria results from reduced resorption of water from excrement in the colon. Ask owners if the feces have the same appearance at home to determine if this is pathologic.

Survey films need to be properly positioned to aid diagnosis. Often, the gastrointestinal tract will be difficult to delineate. It is often possible, particularly in galliformes, waterfowl and some psittacines, to visualize the ventriculus due to the presence of grit. If the proventriculus is dilated, it will bulge past the imaginary lines on the left side on the VD; bulging on the right is usually liver or intestinal.

The shape of the cardiohepatic waist varies by species, so interpret this area carefully.



Proventricular:keel ratio

Normal proventriculus:keel ratio is <0.5 - see radiology paper provided by Dennison et al in optional reading.

Example of proventricular dilation in an African Grey Parrot. This parrot was euthanized a week later due to poor prognosis and progressive clinical signs. The proventriculus:keel ratio measurements are demonstrated by the black lines and the ratio was calculated as 0.61.

Contrast may be used to outline structures within the gastrointestinal tract. Barium can be diluted 50:50 with water or with feeding formula but we generally use it undiluted here at UCD. It may, additionally, have a soothing effect on the GI tract. GI transit times will differ for different species; in several text books there are charts to describe what you could expect in healthy bird species. In this case, contrast can tell you you have a mass effect ventral to the GIT

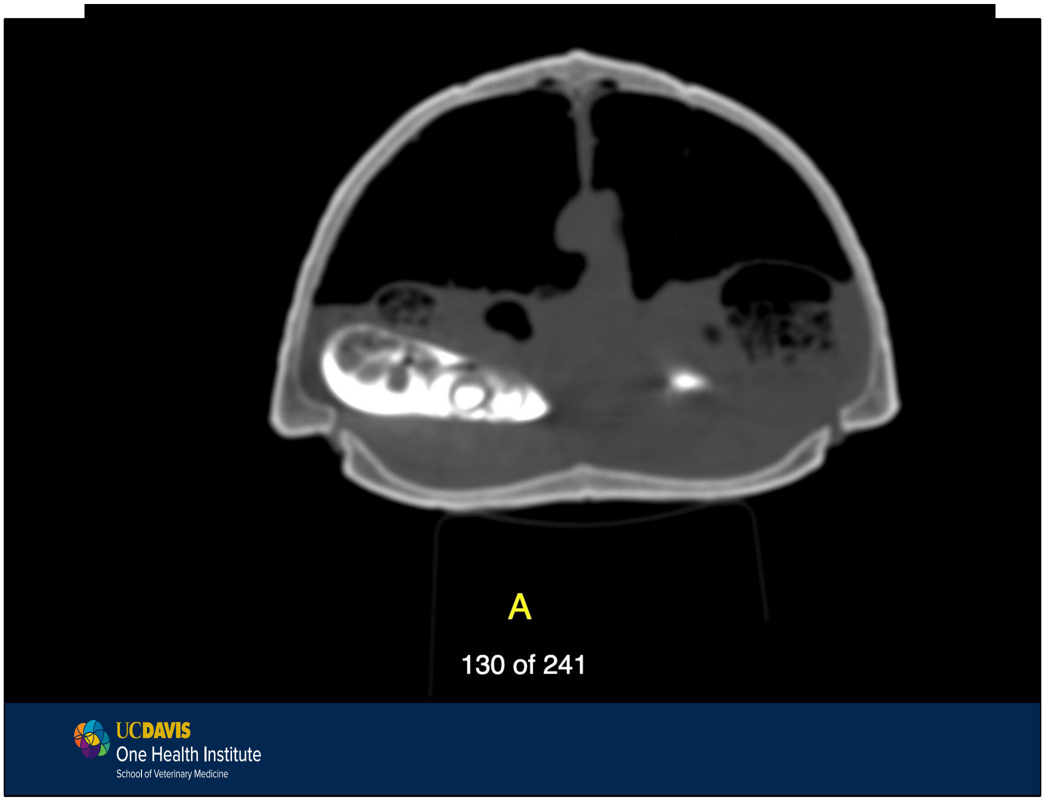
Do not use barium in cases of suspected GI perforation or when planning GI endoscopy and use care to prevent aspiration.

Iohexol may also be used, especially if there is potential for GI perforation. Tends to dilute by the end of the series and poor contrast of lower GI tract. Anesthesia and some sedatives may alter motility.

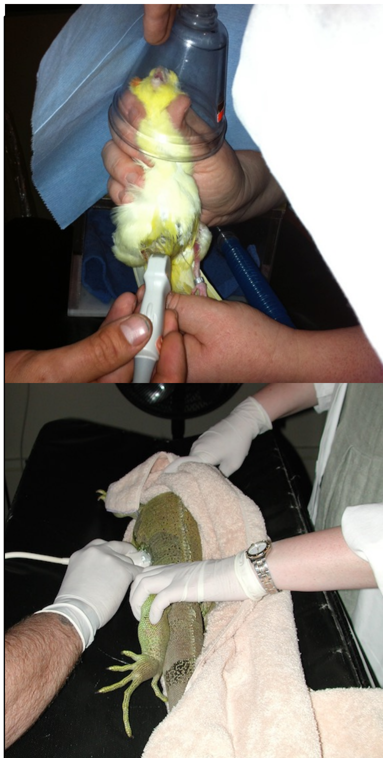
Fluoroscopy is invaluable for evaluating normal motility in birds – normal transit times and ventricular motion rates have been published for Amazon parrots but also now for grey parrots and even for barred owls.

This is the normal fluoroscopy of a grey parrot showing 1-2 ventricular contractions per minute. It is important to know the normal before y0u can understand the abnormal. The contrast in the crop is normal in that it is released slowly as it acts as a holding vat until needed.

Identification of decreased motility used can be used in diagnosis of dilated proventriculus but also in many other diseases of the GI tract such as neoplasia and infections such as Macrorhabdis ornithogaster. While this is still not definitive for any diagnosis, the images provide functional information that contrast or CT cannot provide. Endoscopy is generally not useful, when the proventriculus is so dilated, it becomes dangerously thin and thus may perforate during an endoscopic procedure.



CT can be useful for determining where in the GIT specifically less radiodense foreign bodies might be, or especially in reptiles help us with specific anatomic positioning for surgical planning. Here, CT was used to help determine where in the GIT an ear bud eaten several days before had travelled in a 3-toed box turtle. Of course it was stuck at the pylorus wreaking havoc!



Advanced Imaging

- Ultrasound
  - Hindered somewhat by keel and air sacs of birds
  - Easier when pathology exists
- Endoscopy/Coelioscopy
  - Gastric, esophageal, colonic, cloacal

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Avian ultrasound is possible by using the liver as an acoustic window. In some birds, the air sacs will be more prominent and may hinder visualization. If birds have hepatomegaly or coelomic fluid, ultrasound images will improve. It is normally not possible to distinguish layers of the intestine, but you may identify masses or foreign objects. Motility can also sometimes be assessed.

Reptile ultrasound can be more useful, but a thorough understanding of anatomy for the different species is required by you, the exotics specialist as the radiologist is going to depend on you for the anatomy via textbook, drawings, etc in order to get a full picture of normal versus abnormal.

Outline

Upper GI	Lower GI
• Beak disorders • Infectious disease • Foreign body	• Infectious disease • Foreign body • Neoplasia • Cloacal prolapse

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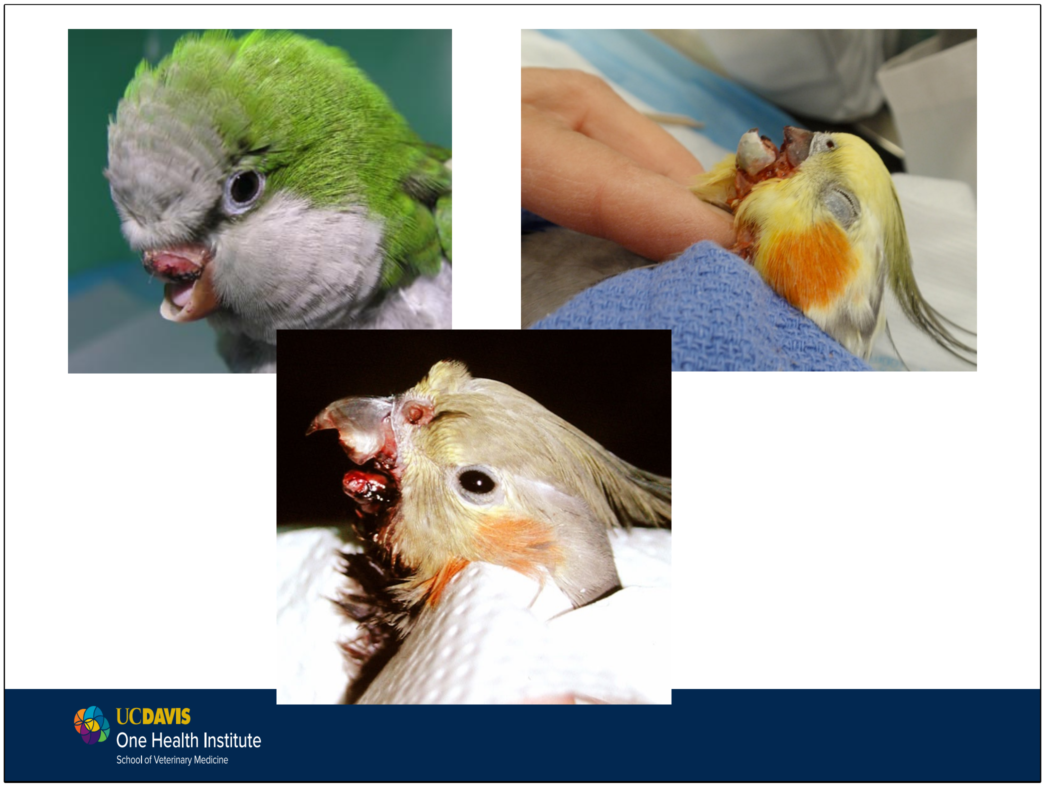
Review your anatomy notes!



It is not uncommon for birds to lose either the gnathotheca or rhinotheca as well as the underlying mandibular or maxillary bone during fights with other birds. The prognosis depends on the severity of the injury, other injuries sustained, the amount of blood loss and the ability of the bird to eat with the injury. If the client really wants to try to save the bird, it is worth a try but daily body weight checks are mandatory to ensure body weight loss isn't too significant as the bird learns a new way to eat. In some cases, an ingluvial feeding tube can be placed to supplement calories, especially in small birds whose metabolism is so high.

Beak Disorders

- Neoplasia
  - Fibrosarcoma
  - SCC
  - Melanomas
  - locally aggressive, difficult to treat
- Infections
  - *Mycobacterium tuberculosis*

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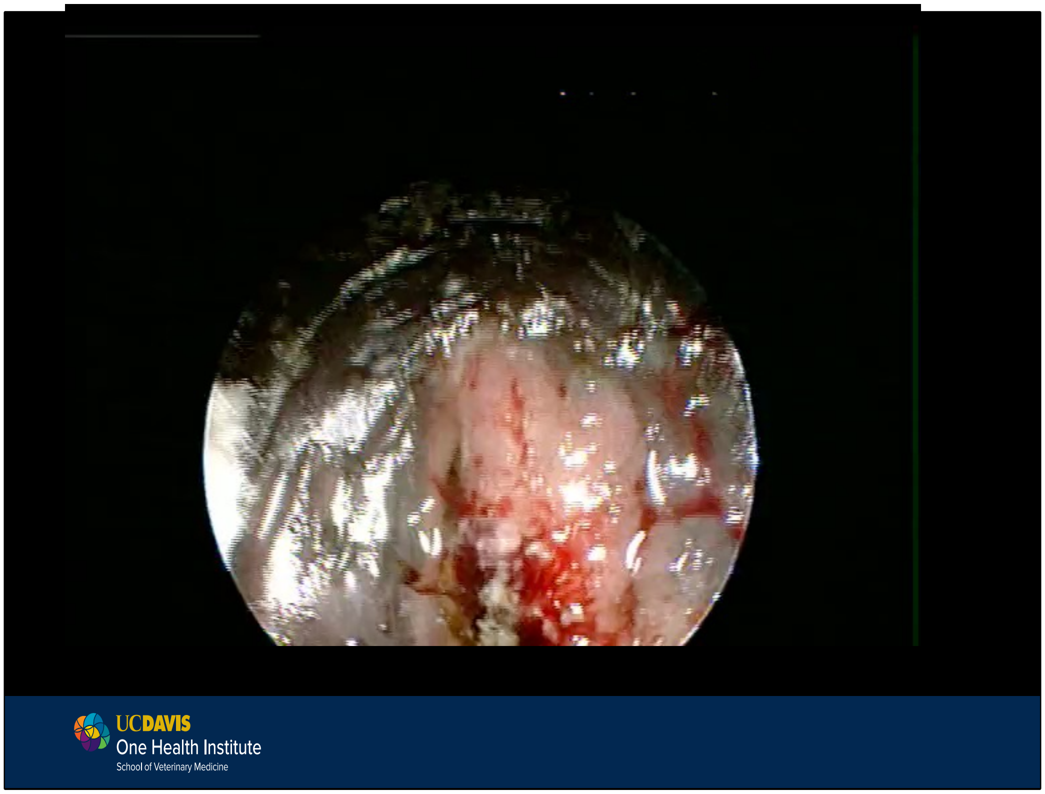
Neoplasia: Fibrosarcoma most common on beak, but SCC, malignant melanoma of the oral mucosa are very common.

These tumors are relatively common in psittacines and are quite locally aggressive. They may also be seen within the casque of hornbills. Treatment with radiation and chemotherapy have been attempted, with variable responses.

Oropharyngeal Plaques or Masses

- Hypovitaminosis A
  - Parrots
  - Better diet, PO vit A
- Trichomoniasis
- Candidiasis
- Capillariasis
- Poxvirus
- Neoplasia

Many lesions look alike (often plaques or raised masses), species of bird and the history may guide differentials and diagnostics.
Be sure to do a wet mount with warm saline and examine within 15 minutes.
Diagnostics to consider: cytology, gram stain, acid fast stain, wet mount, C&S, biopsy and histopath, serum vitamin A levels (in larger birds).
Neoplasia affecting the orophaynx can include squamous cell carcinoma, melanoma or lymphoma



A movie showing an oral squamous cell carcinoma via oral endoscopy in an African grey parrot taken from our clinic. Note the hemorrhage and ulceration typical with these. Dr. Guzman will talk about treatment of these tumors in the surgery lecture.

Trichomoniasis

- *Trichomonas gallinae*
  - Protozoa affecting wild birds predominantly
  - “Canker= pigeons
  - “Frounce”= raptors
  - Decimated house finch population
  - Caseous oral plaques
    - Can extend all through esophagus

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Etiology: *Trichomonas gallinae*
“Canker” in pigeons; “Frounce” in raptors- not a disease found in pet birds except in captive Columbiformes, raptors (falconers birds, etc)
Clinical Signs: Caseous oral plaques
Diagnosis: Wet (saline) swab; crop lavage
Transmission: direct contact
Treatment: Metronidazole; dimetridazole, carnidazole, ronidazole
Control: Sanitation

The flagella and the undulating membrane give this parasite a characteristic quivering motility

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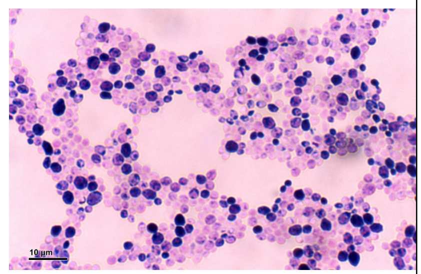
Etiology: *Trichomonas gallinae*
“Canker” in pigeons; “Frounce” in raptors- not a disease found in pet birds commonly except in captive Columbiformes, raptors (falconers birds, etc) where it is very common and those fanciers use the terminology above when describing!
Clinical Signs: Caseous oral plaques
Diagnosis: Wet (saline) swab or crop lavage; “falling leaf” or “herky jerky” motion of the flagella and the undulating membrane give this parasite a characteristic quivering motility that distinguishes it from other protozoa.

Transmission: direct fecal-oral contact
Treatment: Metronidazole; dimetridazole, carnidazole, ronidazole
Control: Sanitation

Candidiasis

- *Candida albicans*
  - Polymorphic yeast, opportunistic
  - Young birds on antibiotics
  - Long term antibiotic therapy
  - Caseous plaques, respiratory
  - Cytology, culture, PCR
  - Anti-yeast medications
    - Fluconazole, ketoconazole, terbinafine



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There are >150 spp of Candida sp. White plaques may be present in the mouth or can extend down the esophagus creating mass effects. Owls in particular come in in respiratory distress from the plaques occluding the glottis or from
Regurgitation, anorexia, and delayed crop emptying may be present with a Candida infection of the crop. Some birds develop a swollen or bloated mucus-filled crop.
Prevention – hygiene
Diagnosis – gram positive budding yeast, culture
Treatment –treatments for YEAST: nystatin, ketoconazole, f**luconazole**, terbenafine

Capillariasis

- Nematode in wild birds, BYP
  - Esophagus, crop, intestines
- Caseous plaques, diarrhea
- Oral cytology, fecal flotation
  - "Caps" on both ends of eggs
- Fenbendazole
  - Caution with many bird species!
    - Pancytopenia, death
    - Use lowest dosage necessary
  - Now using flubendazole, mebendazole, thiabendazole

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Multiple species of capillariasis infect birds. Most prevalent in wild waterfowl and raptors but seeing it now in backyard poultry quite a bit.

May cause anorexia, dysphagia, weight loss, weakness, diarrhea

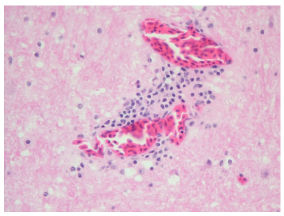
Treatment – fenbendazole, mebendazole, thiabendazole, flubendazole.

Esophagus & Crop

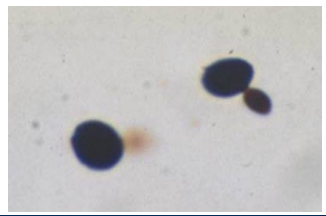
- Infectious

Gram neg bacteria

ABV



Candida



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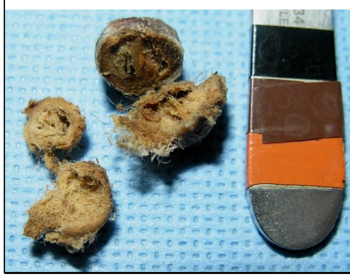
Gram negative bacteria should not been seen in the crop of parrots and often indicate bacterial overgrowth or altered crop/GI motility. Yeast are not identified in the crop and should never be seen budding as this indicates an active infection. Candida sp. may be evident on cytology and is easily isolated.

Avian bornavirus (ABV) can cause crop stasis and in fact the biopsy of choice is the crop and a lymphoplasmacytic ganglioneuritis is the lesion. Will discuss in detail in proventricular section. Image from vet.uga.edu. Scarlet Macaw, crop, H&E stain. Severe lymphoplasmacytic ganglioneuritis in experimental PDD.

Esophagus & Crop

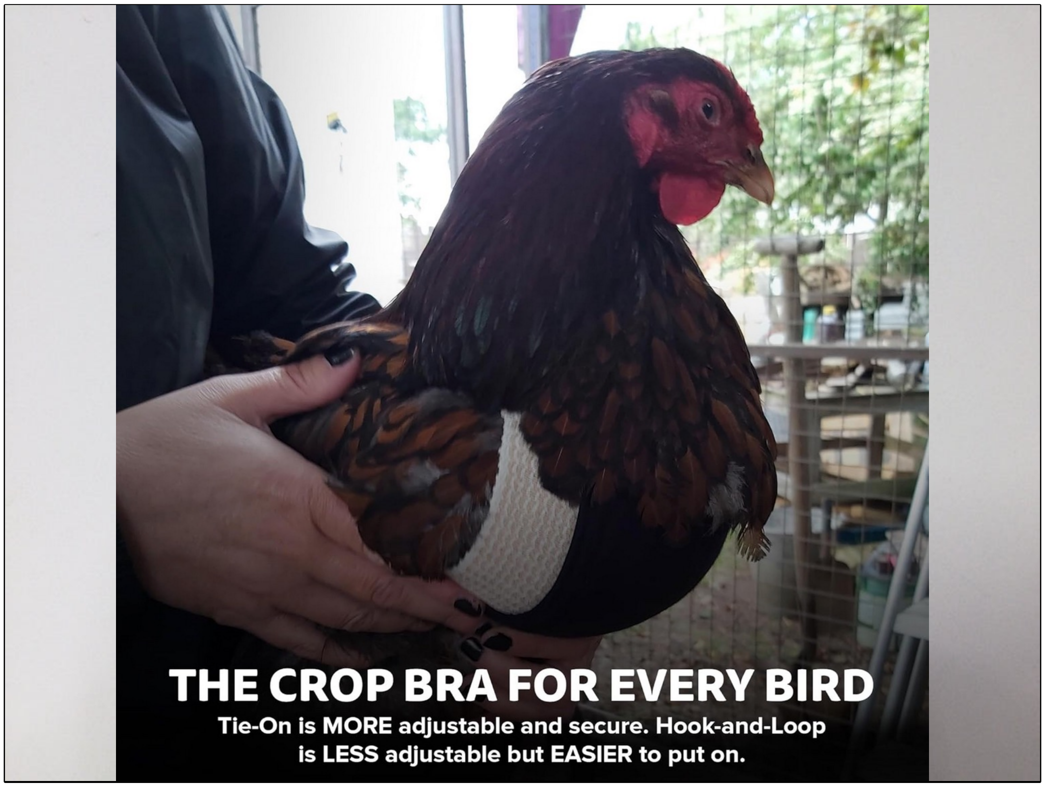
- Crop stasis
  - Feeding too large volume
  - Food impaction (ingluviolith)
    - Dehydration, “spring grass” in BYP
  - Toxicosis
  - Foreign bodies
    - Hardware disease in BYP



Foreign bodies are especially common in unweaned or juvenile birds but hardware disease and heavy metal toxicosis is very common in BYP. Common FBs in pet birds include cage substrate, grit, or toys, but may be secondary to ingluvioliths, heavy metal ingestion, or eating large food items that are difficult to pass beyond the crop. Clinical signs include vomiting or regurgitation, decreased feces, anorexia, weight loss, lethargy. Foreign body removal can be performed by endoscopy, ingluviotomy or crop lavage with a red rubber catheter.

Excess crop distension can occur due to feeding large volumes. Crop support (crop bra) may be needed to return the crop to a more normal shape. In older birds, it may occur with general enteritis, heavy metal toxicosis, foreign bodies or feed impactions. For example, BYP often released onto “spring new growth grass” will gorge and create an impaction.

Ingluvioliths are uncommon but easily retrieved. An underlying cause, dehydration, improper food materials (gravel, corncob, grit), should be investigated for ingluvioliths. Enlarged thyroid secondary to goiter may occur in animals on all seed diets, but were more common in budgies. These birds may be dyspneic if the thyroid compresses the trachea.



Crop bras are not just for cockatoo chicks anymore! For chickens or any bird that has a pendulous crop, “crop bras” are now marketed just about everywhere to help keep the crop in its place. Make sure the fit is correct, I’ve seen some that do not do the job correctly. Many are sold now for chickens as they are the major market for them, but many of these can be scaled in size, or simply use vet wrap and make your own for cockatoo chicks needing a more temporary solution.

Esophagus & Crop

- Iatrogenic burns, fistulas
    - Especially in youngsters
  - overheated formula (>110 F)
  - Feeding tube injuries
  - Delay surgical closure until burn has "declared"
    - Often 3-5 days

Surgical closure

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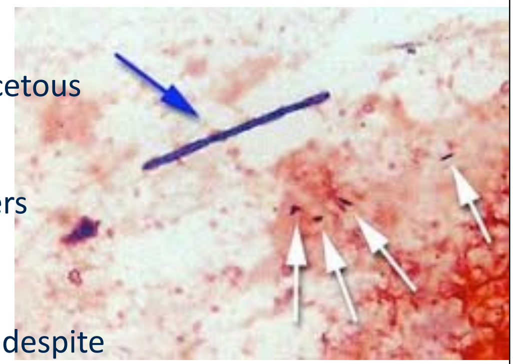
Iatrogenic laceration of the esophagus or crop can occur following the use of a rigid feeding tube. Crop burns/fistula: This normally occurs in unweaned psittacines fed formulas that are too hot. Microwaving formula is often the cause due to uneven heating. Damage to the crop may interfere with crop motility. Mild burns may cause inflammation and edema that may resolve. Birds will present with voracious appetite and weight loss. On physical examination, moistened feathers are noted in the area of the crop, normally a scab is present. It is recommended to wait 3-5 days in the case of full thickness fistula. The tissue needs time to show extent of damage, otherwise repair may fail. Extensive burns require removal of extensive area of crop, prognosis may be poor due to lack of normal function.

Iatrogenic laceration of the esophagus or crop can occur following the use of a rigid feeding tube.

Proventriculus & Ventriculus

- *Macrorhabdus ornithogaster*
- “Avian Gastric Yeast”
  - very large anamorphic ascomycetous yeast
  - Cockatiels, lovebirds, budgies, parrotlets, chickens, many others
  - Lives in “isthmus” of stomachs
- Clinical Signs
  - regurgitation, weight loss, ADR despite ravenous appetite (or anorexia), undigested food in feces



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Also called ‘megabacterium” by the old school and “avian gastric yeast” for ease of explaining to clients.

It has been found in so many species now, zoo collections and many new reports from Middle East. We used to think of little parrots as the primary species, but now it seems most species can be infected. The incidence in normal birds is not known and why it sometimes causes disease an sometimes it doesn’t.

Clinical signs include lethargy, weight loss, emaciation, anorexia, regurgitation, vomiting, dark feces, passing undigested seeds or other foods. Grossly, there may be distention of proventriculus, mucus buildup of mucosal layer of proventriculus, thickening and hemorrhaging of the proventricular wall. (Image from Y. Hannafusa, JVDI 2007. Hemotoxylin and eosin–stained section of the koilin layer of a 7-day-old chicken infected with *M. ornithogaster*.)

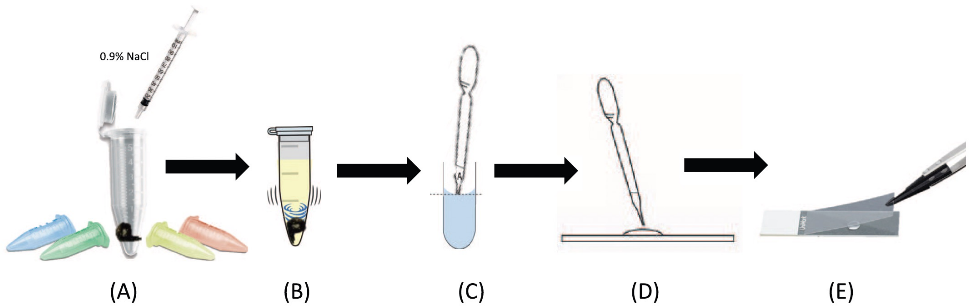

Figure 1. A pictorial representation of the macro fecal suspension technique: (A) 0.3 mL of 0.9% sodium chloride is placed in a microcentrifuge tube with 0.5 g of feces; care must be taken to avoid contaminating the suspension with the uric acid portion of the droppings. (B) The lid is closed and a slurry formed by shaking the microcentrifuge tube manually for 20 seconds. (C) The slurry is placed in an upright position for 3 seconds to allow particulate matter to settle before removing 0.06 mL from the meniscus with a 1-mL syringe. (D) A drop is placed on a microscope slide and (E) covered with a coverslip for analysis.

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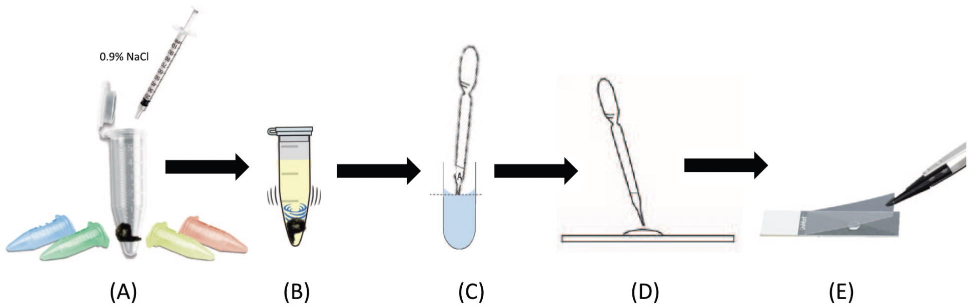
Diagnosis has been in the past using either Gram stain or PAS stain on feces alone. Now a Master's student in Australia evaluated this and determined that using the " macro fecal suspension" technique provided superior results- if one can obtain 0.5 gm of fecal material for the test. Adding Gram's stain didn't help, the organisms moved off to the periphery.

There is a recent paper showing not only resistance to clearance of AGY, but a rebound effect occurring in some birds after treatment.

BARON ET AL—*MACRORHABDUS ORNITHOGASTER* DIAGNOSTIC MODALITIES 39



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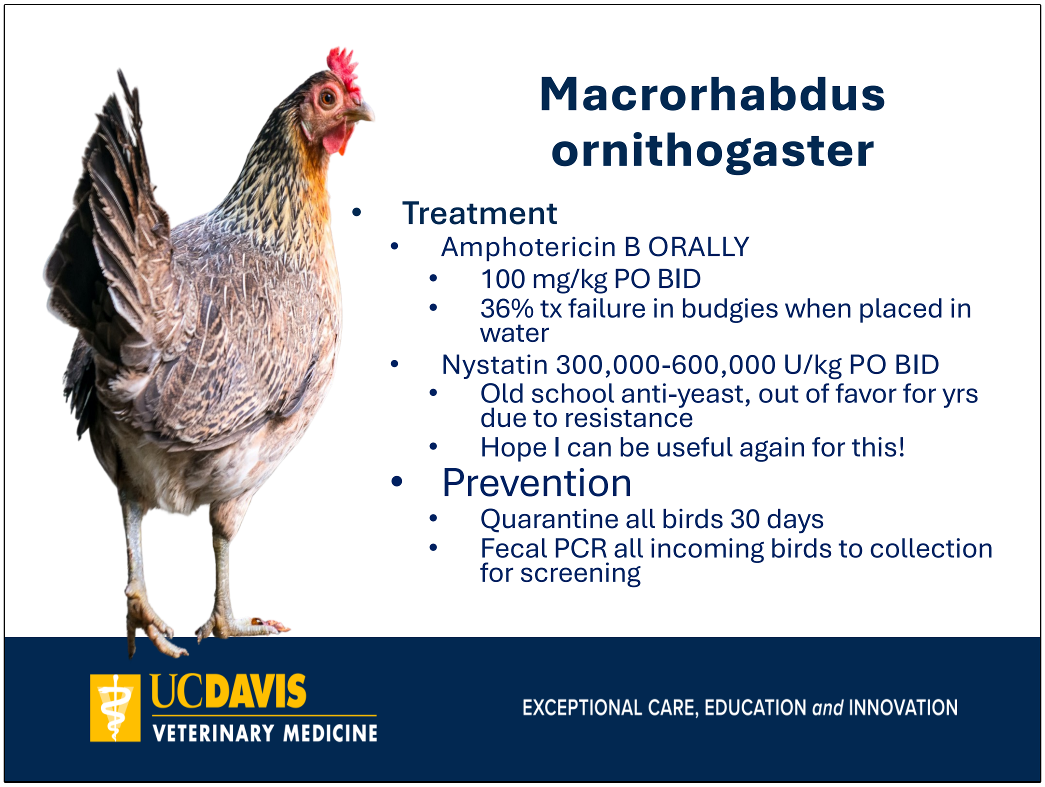


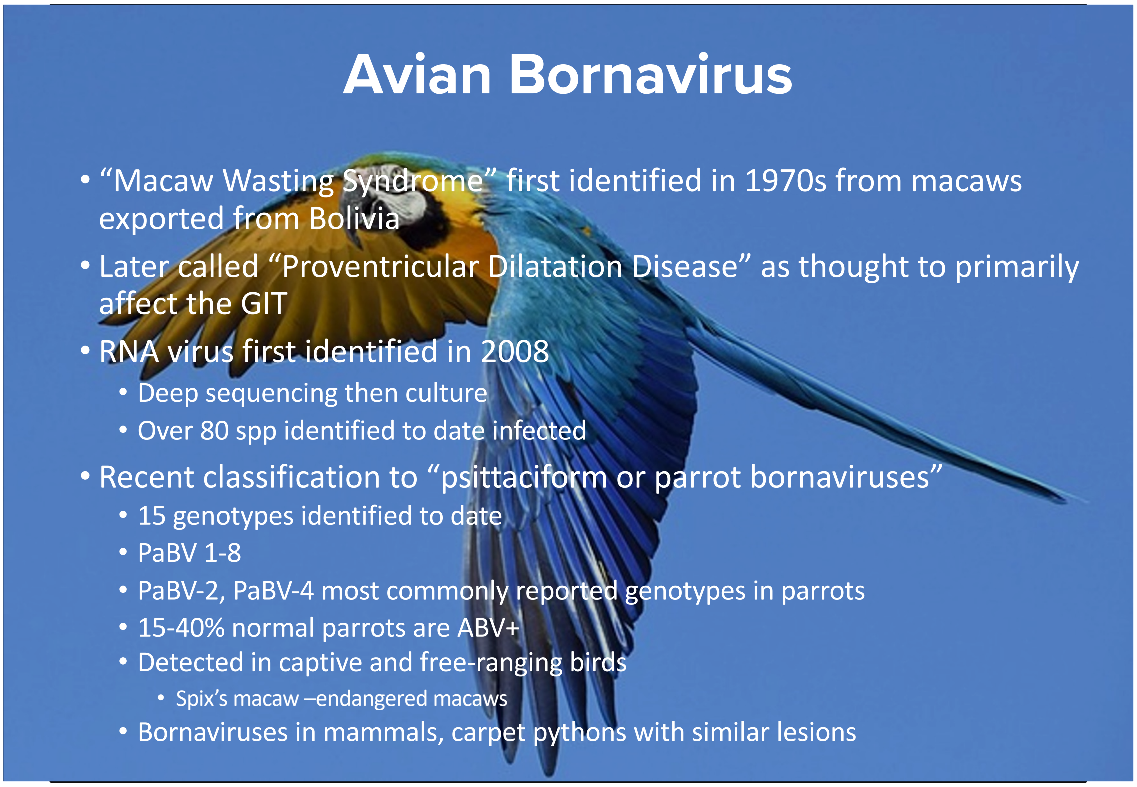
Treatment of M ornithogaster is very challenging. Amphotericin B ORALLY has been effective in very few situations but has been more successful than just about every other antifungal attempted. This drug is not absorbed orally so little chance for toxicity when given by this route. However it also has not always cured the patient either, very worrisome given we have very few fungicides (voriconazole at high doses is fungicidal but also can be toxic).

In very small birds in aviaries, placing water soluble amphotericin B in the drinking water has been used, somewhat effectively. Be aware that budgerigars are difficult to medicate in the water with any medication because they are desert species and drink very little water, making this route challenging for any drug. So when a study came out showing that 36% failure occurred in a budgie aviary no one should have been surprised. But recheck pooled samples from aviaries after water treatment to ensure clearance.

Virtually every antifungal has been attempted with little success. As it is a yeast, fluconazole at 100 mg/kg was found to be somewhat effective but at such a high dose these is significant danger of toxicity.

Nystatin is a very old anti-yeast out of favor for decades due to resistance. It works directly on the GI tract, like ampho B it does not get absorbed. Some clinicians have used at 300,000-600,000 U/kg and have found it to be effective. With little toxicity, there is hope this drug may be effective against this organism.

Prevention best is quarantine and screen- screen before accepting birds to your flock if possible. If not, PCR feces at least twice over 30 days.



<u>**MY OPINION THIS IS THE MOST IMPORTANT GI DISEASE OF PSITTACINES CURRENTLY****</u>**

• Bornaviruses are negative-strand RNA viruses in family Bornaviridae. PDD has been reported to occur in more than 80 species of psittacines. It has also been reported in toucans, honeycreepers, weaver finches, waterfowl, raptors, and passerines. PDD is a fatal neurologic disease that uniquely affects the nervous system, specifically the enteric nervous system. The disease was first recognized in the early 1970s in macaws exported to Europe, N.A. from Bolivia. The infection has spread worldwide now. • Because this was an unknown syndrome for over 30 years, it had many names- macaw wasting syndrome, PDD, ABV, and now many are requesting to call it "avian ganglioneuritis" • High fatality when clinical signs are present • In 2008 deep sequencing of DNA from brains of parrots with PDD identified 2 strains of a novel bornavirus. Then used real-time PCR to confirm ABV in brain, proventriculus, adrenal gland of a number of birds. At the same time, another group succeeded in culturing ABV from brains of seven parrots with PDD which provided a source of antigen for serologic assays and nucleic acid for molecular assays

• In 2012, a German group infected cockatiels intracerebrally with ABV-2 and saw some clinical disease including neurological disease but not in all the birds, whereas a few years later they conducted the same study using ABV-4 and found that birds showed more GI signs with this strainthan did the birds with the other strain, suggesting that a range of signs can be seen with differing strains of virus in differing species as well.

- In 2014 investigators found PaBV-4 in free-ranging macaws in Brazil, so this is not just a disease of captive birds.

• The recent reclassification of ABVs has identified 15 genotypes within 6 species to date.
    - Psittaciform bornavirus 1 and 2 are the two species known to cause clinical signs of avian ganglioneuritis (which includes parrot bornavirus [PaBV] 1-8 genotypes)
    - PaBV-2 and PaBV-4 are the most common genotypes reported in parrots
    - Other avian bornaviruses include:
        - Passeriform bornavirus 1 (CnBV-1, 2, 3, canary) and (MuBV-1, munia finch), waterbird bornavirus 1 (ABBV-1, aquatic bird bornavirus 1), passeriform bornavirus 2 (EsBV-1,estrildid finch bornavirus 1), and tentative, unclassified bornaviruses of ABV-MALL, parrot bornavirus5 PaBV-5, parrot bornavirus 6 PaBV-6, and parrot bornavirus 8 PaBV-8.
- Whether a virus reservoir such as known for the mammalian bornaviruses exists and whether wild birds play a role in this scenario needs to be further investigated.
- Jungle carpet python virus (JCPV) and Southwest carpet python virus (SWCPV) of the genus Carbovirus were discovered by metagenomic and PCR screening in Australian pythons (Hyndman et al., 2018).
- Virus detection was variably associated with neurological signs such as progressive caudal paresis, flaccid paralysis, delayed righting reflex and incoordination. Histological lesions consisted of non-purulent encephalitis and ganglioneuritis. Detection of virus without neurological signs or histopathological lesions is similar to what has been found in birds.

Update on Avian Bornavirus and Proventricular Dilatation Disease

Diagnostics, Pathology, Prevalence, and Control

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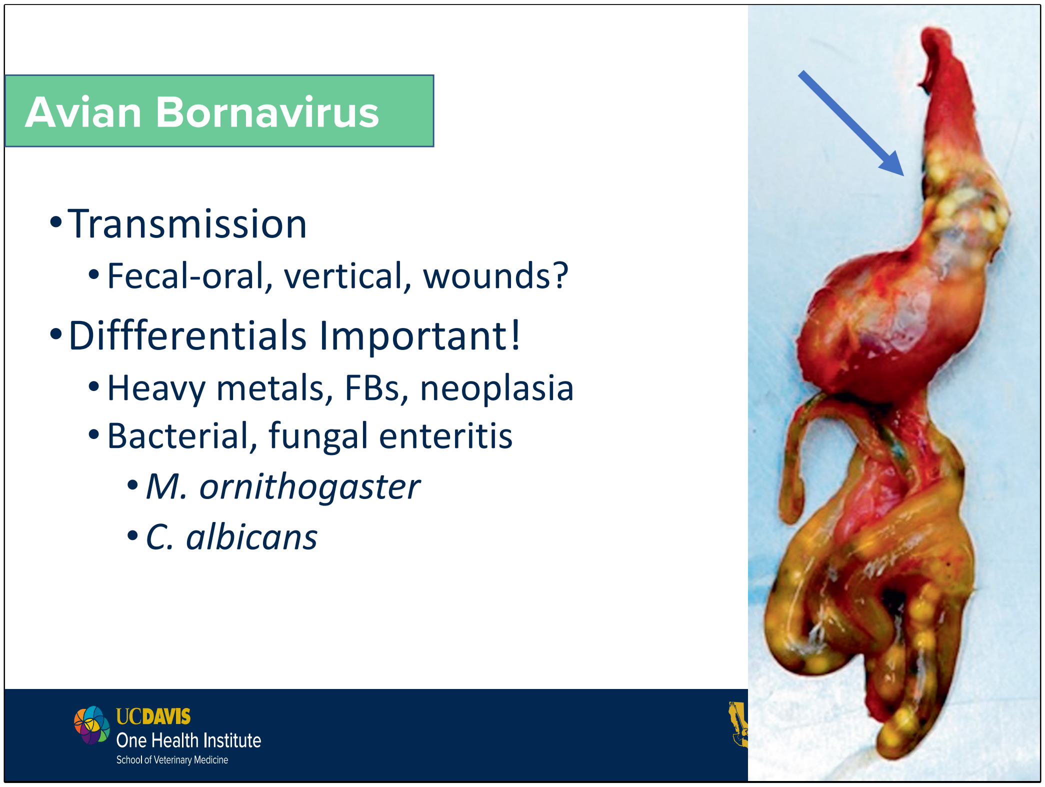
KEYWORDS

• Avian ganglioneuritis • Avian bornavirus • Proventricular dilatation disease
• Bornaviridae

KEY POINTS

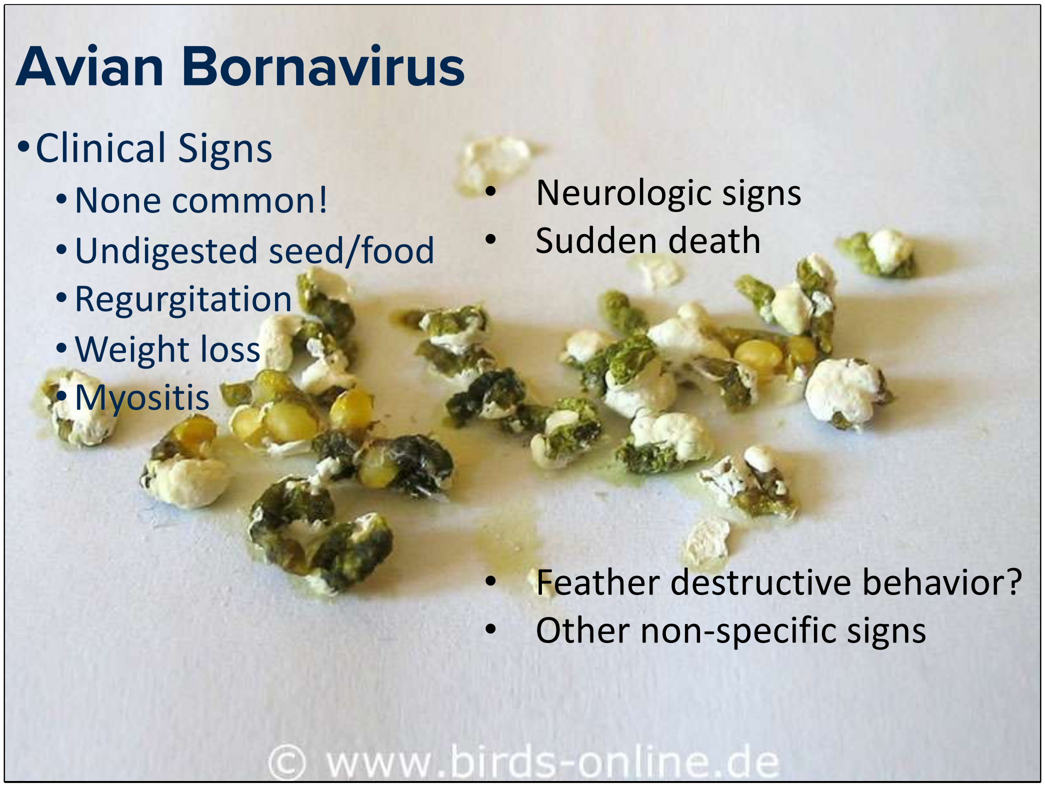
- Avian bornavirus (ABV) is a causative agent of proventricular dilatation disease in birds.
- Birds intermittently shed the virus in their droppings, and seroconversion is unpredictable. This makes diagnosis challenging, requiring multiple negative polymerase chain reaction (PCR) and serology tests before determining that a bird is negative.
- Different genotypes of ABV may be more or less virulent depending on species of bird infected. Having exposure to one strain may not provide immunity to another strain.
- There is a large percentage of exposed birds that are serologically and/or ABV PCR positive yet clinically normal. It is unknown if these birds will ever develop disease.
- A clinically normal ABV-positive bird should not be euthanized.

This is a 2020 VCNA paper in your folder that is as current a review that covers diagnosis and treatment as there is now. You will hear many more syndrome names for this disease- avian ganglioneuritis is the term for the lesions whereas the virus is avian bornavirus. A 2022 review digs deeper into the pathogenesis.



- Transmission is still uncertain but there is ample evidence that the disease is contagious. The virus has been found in feces, so fecal-oral is considered the most plausible. But it has also been found in urine, the respiratory system and in feather calami.
    - Cockatiels did not become infected when exposed orally or nasally
    - Vertical transmission occurs in mammals with bornavirus infections, but hasn’t yet been demonstrated in parrots.
    - Host factors such as species, age, and immunosuppression may play a factor
    - Some have proposed that there may be a need for presence of mucosal or skin lesions for infection to occur
    - Large number of unaffected positive birds makes incubation periods challenging to determine. Five week old chicks exposed to a sick parrot with PaBV died within days whereas adult birds died within weeks after acute infection. But decades may go by before a bird living without other birds have developed clinical signs.
    - Neurologic dysfunction occurs throughout the neuroganglia of the GI tract. The intestinal dysfunction is probably due to virus-induced immune damage to the autonomic nerves affecting the upper and middle GI tract. The disease is also associated with significant central nervous system damage.

- Image of cockatiel GI tract showing thin wall of proventriculus from. Speer 2016 Current Therapy

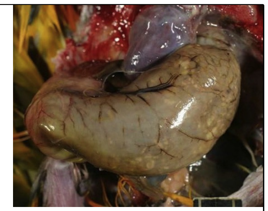


- Clinical signs range from weight loss, crop stasis, proventricular and intestinal dilatation, regurgitation, and maldigestion to eventually starvation and death.

- Signs involving the CNS may also be present, such as tremors, ataxia, seizures, and blindness. The myenteric plexi of the heart and adrenal glands have also been affected. Birds may have GI signs, CNS signs or both.

- Feather damaging behavior (FDB) has been associated with PDD, but scientific evidence for causality is lacking. A recent article by Fluck et al. evaluated viral RNA shedding and PaBV antibody titers in clinically normal birds, birds with neurologic signs, and birds with FDB. Birds with neurologic signs had the PaBV highest titers and highest viral shedding, followed by birds with FDB having significantly lower values and clinically normal birds having the lowest to no PaBV titers. There was no significant difference between antibody titers in the birds with feather destructive behavior and the clinically normal birds. This study provided no evidence to support PaBV causing feather damaging behavior, but further studies are necessary to rule out any link between PaBV and FDB. ABV has been detected in the skin.

Avian Bornavirus



- Antemortem Diagnosis
  - Commercially available RT-PCR, serology
    - Antigenic diversity, sporadic viral shedding make testing challenging
    - Must use established laboratory but PCR techniques not standardized
    - Urine, feces, choanal/cloacal swab best? Blood, feces often negative.
    - If suspect, may need repeat testing
    - PCR of feather found + even after several years storage
      - Can be contaminated from other birds if feather only
    - Titers only show exposure; several different tests available now
      - Antibodies cross-react with many genotypes, but not all?

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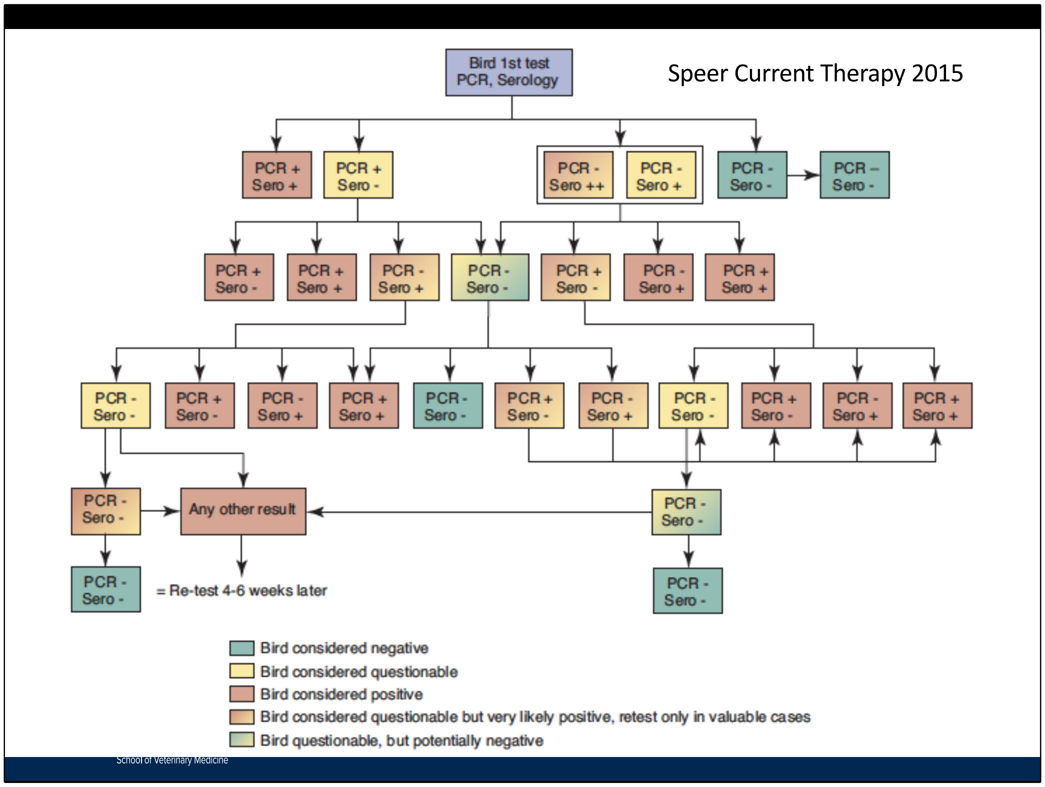
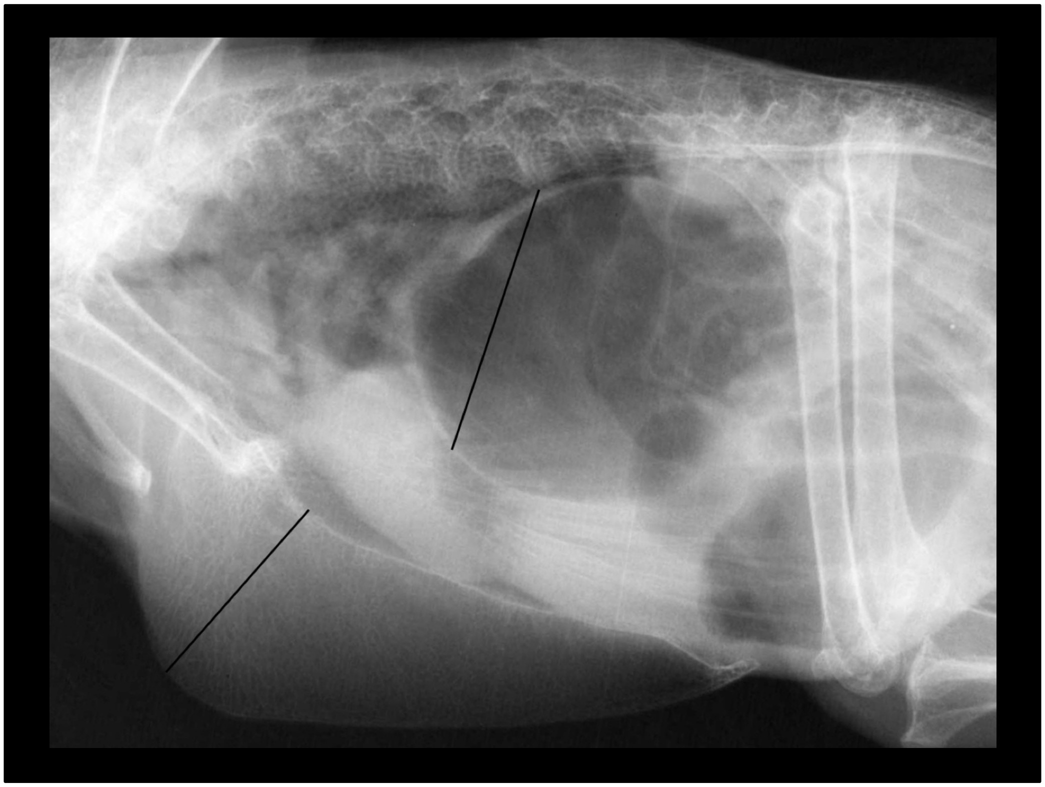
PCR
- Multiple diagnostic tests available but interpretation of PCR positive is difficult if birds may be infected without evidence of disease and viral shedding may be sporadic.
- RT-PCR may be insensitive due to the many genotypes. Birds mount an antibody response to bornavirus, but normal healthy birds can also have antibodies to ABV- which does not predict disease consistently.
- Urine, feces, and cloacal swabs are the samples most likely to contain virus.
  - Shedding, however, is intermittent, which can lead to a false-negative test result. Many healthy infected birds shed ABV infrequently and conversely many confirmed cases of PDD have been reportedly PaBV negative with multiple tests.
- Some birds, especially those with clinical disease, may shed the virus on a continuous basis.
- Because of all these variations, the testing of a single dropping or cloacal swab from a single bird is of limited usefulness.
- deKloet and colleagues suggested RT-PCR performed on a feather sample is a reliable testing procedure. However, others have had limited success using the RT-PCR assay on feather calami, and the opinion among most other researchers was that the feather PCR was not an adequate sample for accurate testing. Similarly, Dalhausen and Orosz suggested that the use of RT PCR on whole blood of birds is a sensitive diagnostic test, but other studies have not supported this.
- Although RT-PCR is the most commonly performed test for ABV, at this time there is no standardized method of testing among laboratories offering this service.
- Reported rates of ABV positive birds among commercial laboratories and university settings range from 3% to 33%.
- This may be due to intermittent shedding or because the known ABV genotypes vary in sequence identity so some assays may not detect all genotypes.

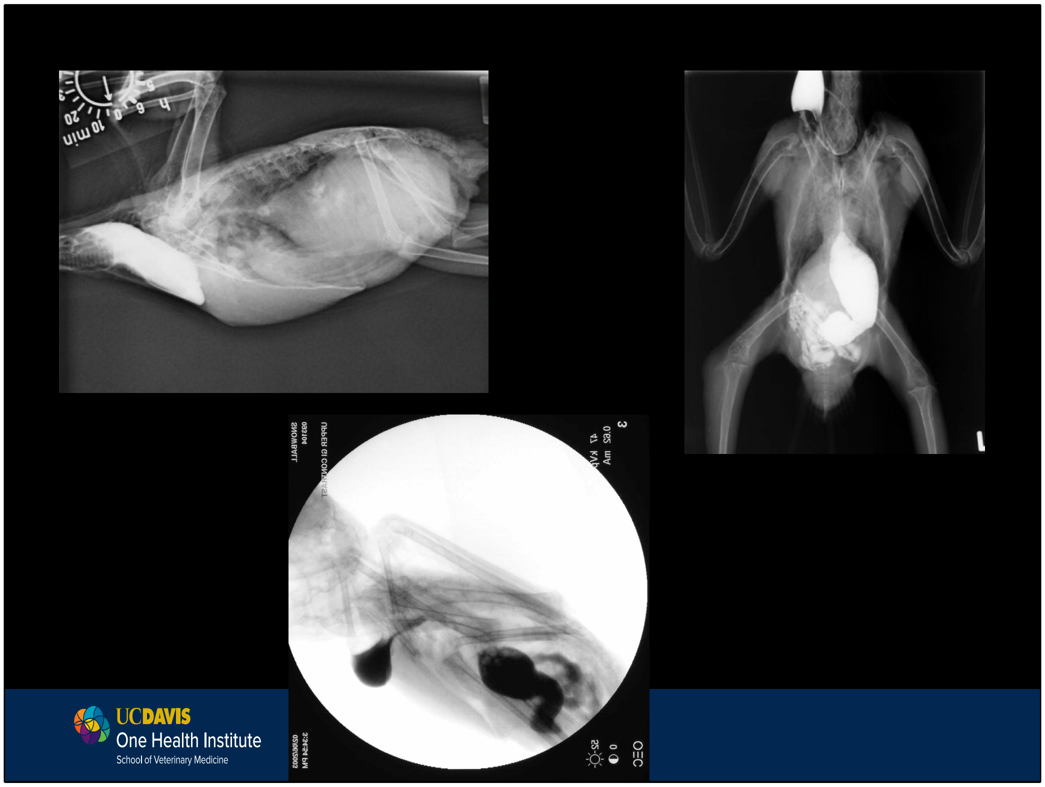
Serology
- Although PaBV infection is common, the development of clinical PDD is rare. Tests that simply detect the presence of PaBV may not be useful for the antemortem diagnosis of PDD.
- This is further complicated by some PaBV-infected birds not developing a detectable antibody response, although, conversely, there does seem to be a positive correlation



This table is from Dr. Speer's 2016 textbook – a helpful algorithm for determining best practices in testing birds for this disease as of this course.

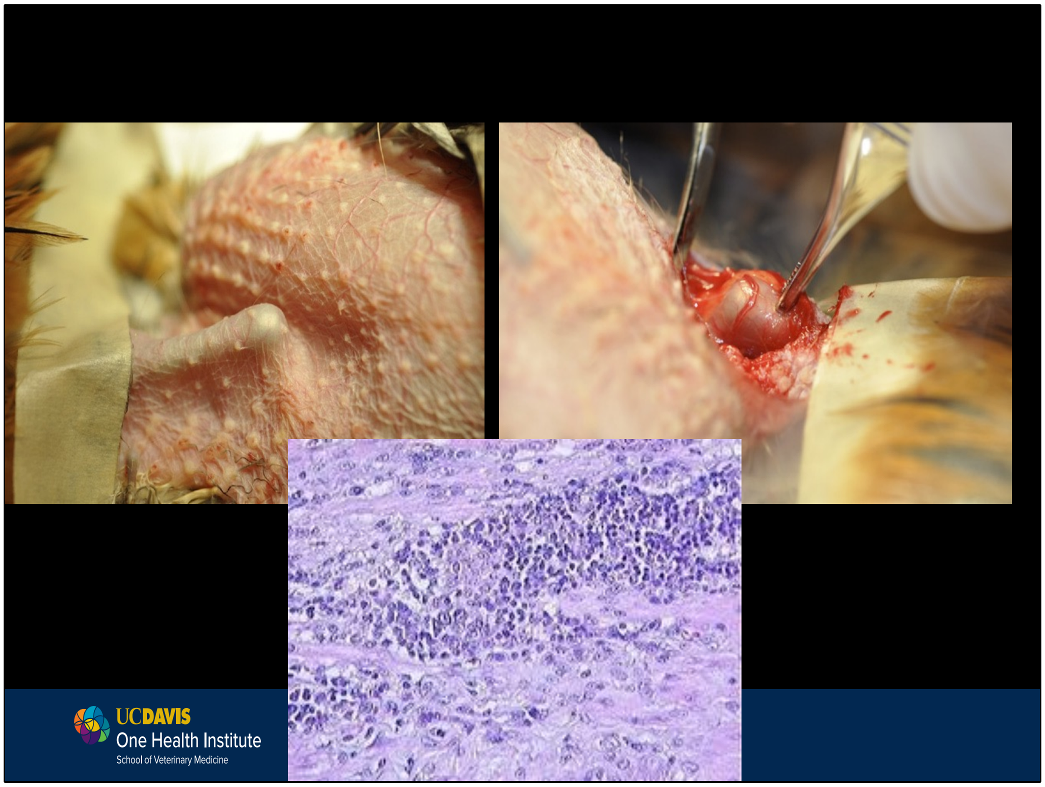


- Other diseases can also cause PDD so don’t condemn the bird to this diseases until other ddx have been ruled out.
- Example of a cockatoo with foreign body ingestion involving the proventriculus, ventriculus and duodenum. Note gas distention of all 3 organs. The foreign body was successfully removed by surgery from the duodenum.
- The proventriculus:keel ratio measurements are demonstrated and the ratio was calculated as 1.37 but remember many things can cause this—don’t give the bird a death sentence for PaBV until proven!



- Imaging /fluoroscopy – identification of decreased motility used in diagnosis as well as a dilated proventriculus.
- While this is still not definitive for PDD, fluoroscopy images provide functional information that contrast or CT cannot provide. In general 1-2 ventricular contractions per minute is normal for most parrots.

• Endoscopy is generally not useful, when the proventriculus is so dilated, it becomes more dangerously thin and thus may perforate during the procedure.
• Enlarged proventriculus on imaging is supportive but not pathognomonic!



- **<u>The primary lesion with this disease is lymphoycytic plasmacytic ganglioneuritis</u>**
- Collecting a crop biopsy has historically been thought to be the most sensitive diagnostic to confirm a positive bird with clinical signs. Biopsy should contain a blood vessel to increase the chance at getting a nerve (~70% sensitivity). Diagnosis is confirmed with classic histopathologic changes (lymphoplasmacytic ganglioneuritis of ganglia and myenteric plexus of GIT or CNS. Lesions can be inconsistent.
    - Now we know that the sensitivity is much lower- more like 35% sensitivity. (Ouyang, 2009) so failure to recognize lesions is a common reason for diagnostic failure.
    - There are immunohistochemical stains available now to increase this sensitivity!!!
- Ganglia of the crop, proventriculus, ventriculus, duodenum, heart, adrenal gland, spinal cord, brain, sciatic nerve, brachial nerve, vagus nerve are common sites to review on post-mortem examination.

Avian Bornavirus

- Treatment
  - Supportive care only
    - Highly digestible diet, keep hydrated, minimize stress
    - NSAIDs
      - Celecoxib 15-40 mg/kg PO q12h!
      - Robenacoxib 2-10 mg/kg IM weekly q4wks, then monthly
    - Cyclosporine A?
    - Gabapentin? 10-25 mg/kg PO q12h
- Prevention
  - Epidemiology still uncertain
  - Minimize exposure to known birds
  - Not stable in the environment

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Treatment:

- No cure to date.
- Supportive care such as easily digestible food, **high dose NSAIDs** may help decrease inflammation associated with neuro dysfunction. They must be given for life.
  - Celecoxib at high doses currently used most clinically to reduce clinical signs and improve body condition.
  - Robenacoxib IM is becoming a treatment to watch since it can be injected once per week initially then monthly. Biggest thing is don't forget and miss treatment.
  - Cyclosporine A was able to confer protection to cockatiels infected with PaBV but has't been used clinically to my knowledge.

Prevention

- Guarded-grave prognosis once clinical signs identified.
- T-cell mediated, so concerns about whether a vaccine could be effective or even exacerbate clinical signs, but there are some promising studies looking at experimental vaccines so stay tuned!
- Infection with one genotype does not seem to confer protection for a different genotype.
- In the absence of effective treatments and effective vaccines, control must be sought by other methods.
- Fortunately, ABV is not an environmentally stable virus. It is easily destroyed with soap or diluted bleach and is inactivated by ultraviolet light and desiccation.
- Control in aviaries and homes, therefore, requires a multimodal approach.
  - Good hygiene and housing birds outdoors when temperature permits.
  - Avoid overcrowding and provide good nutrition.
  - Isolation of all new, sick, or PaBV-positive birds, separation of birds from a flock when they are PaBV positive.
  - Existing birds and newly presenting birds should be tested using both serology and PCR, and birds should be separated based on the results of this testing.
  - There should be at least 3 negative tests at 4- to 6-week intervals before considering a bird negative.
- With the likelihood of vertical transmission, chicks from infected birds should be incubator hatched, hand raised, kept separate from other noninfected birds, and monitored for development of disease.
- Because of the intermittent shedding and inconclusive serology results, testing and separating birds may require years to obtain PaBV-negative aviaries.
- Healthy birds that test positive with either PCR or serology testing should not be euthanized!! They may never develop clinical signs, or might not for decades. But should treat a PaBV+ bird like a FIV+ cat- do not bring other birds into the house.

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Studies on immunity and immunopathogenesis of parrot bornaviral disease in cockatiels

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ABSTRACT

We have demonstrated that vaccination of cockatiels (Nymphicus hollandicus) with killed parrot bornavirus (PaBV) plus recombinant PaBV-4 nucleoprotein (N) in alum was protective against disease in birds challenged with a virulent bornavirus isolate (PaBV-2). Unvaccinated birds, as well as birds vaccinated after challenge, developed gross and histologic lesions typical of proventricular dilatation disease (PDD). There was no evidence that vaccination either before or after challenge made the infection more severe. Birds vaccinated prior to challenge largely remained free of disease, despite the persistence of the virus in many organs. Similar results were obtained when recombinant N, in alum, was used for vaccination. In some rodent models, Borna disease is immune mediated thus we did an additional study whereby cyclosporine A was administered to unvaccinated birds starting 1 day prior to challenge. This treatment also conferred complete protection from disease, but not infection.

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- Bornaviruses are known to beT-cell mediated in mammals, so there were concerns about whether a vaccine could be effective or even exacerbate clinical signs , but there are some promising results looking at experimental vaccines in cockatiels.
- Hameed et al. 2018 infected cockatiels and found that while birds were still able to be infected with PaBV, they did not get sick with clinical disease. There was mild viral shedding though.
- Stay tuned more work underway!

Proventriculus & Ventriculus

- Heavy metal toxicosis
  - Lead sources – paint, wine foil, toys, shotgun pellets, stained glass, mirrors
  - Zinc – galvanized metal, soft rubber toys, pennies minted before 1982, Monopoly pieces

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• Psittacines tend to chew on anything, get acute lead or zinc toxicosis.
• Examples include galvanized cage wire, lead paint, curtain weights, stained glass windows, jewelry), wine bottle foil, toys, mirror backing.
• Waterfowl tend to ingest metal while feeding (lead shot resembles gravel).
• Raptors (esp. eagles, condors) get lead toxicity from scavenging carcasses killed with lead ammunition. California just recently passed a statewide ban on all lead ammunition sales and use.
• Birds in zoological collections can have problems with coins (pennies since 1982). Mucosal lining of the ventriculus becomes irritated, koilin lining may be damaged irreparably.

- One of the very important new areas of heavy metal toxicosis we as clinicians are facing is in backyard chicken flocks. (see optional reading folder)
- If you don’t look, you don’t know…..
- Lead may be stored within the bones, and levels may elevate in female birds that are forming eggs.

Image from homesteadfowl.com

Proventriculus & Ventriculus

- Zinc in all of these!!

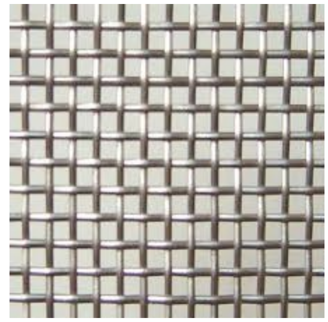

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Image from jectport.ods.org

Proventriculus & Ventriculus

- Heavy metal toxicosis
  - Regurgitation, green feces, crop stasis, pu/pd, neurologic signs, respiratory signs in chickens!
  - Metal object on radiographs, elevated plasma concentrations
  - Zn does not store in body, but Pb stores in bone
  - Treatment
    - Abatement = removal of metal important
      - Surgical or endoscopic removal
      - Medical management with peanut butter, fiber, grit
    - Chelation = Ca disodium EDTA, DMSA
      - Caution with DMSA dosing- narrow therapeutic window

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- Diagnosis may be suggestive based on history, clinical signs, metal FB in GI tract, anemia and erythrocyte abnormalities. Definitive diagnosis is based on elevated blood lead or zinc level. One-shot radiographs can be taken without sedation in ill-birds and may provide a presumptive diagnosis of heavy metal intoxication. Even very small metallic pieces can cause clinical signs. Bright green feces or urates are suggestive, but not pathognomonic, for heavy metal intoxication. The excretion of green pigments is due to build-up of biliverdin. Amazon parrots affected by lead toxicosis may pass pink urates due to renal hemorrhage, the cause for this is unknown.

- Treatment: Acute cases respond well to chelation with calcium EDTA, concurrent fluid therapy should be administered. Attempt to remove metal in GI tract with bulking agents such as mineral oil, peanut butter, or Metamucil or even using "grit." Large objects risk perforation. May require endoscopic or surgical removal.
- Remember lead may be stored within the bones, so Pb levels may elevate in female birds that are forming eggs.
- Chronic or intermittent chelation with oral chelators may be required.
- Zinc is not stored in the body to a significant degree and chelation can often be stopped after the metal has been removed from the GI tract.
- DMSA= Dimercaptosuccinic acid caution should be exercised with DMSA as there is a narrow therapeutic index and deaths have been attributed to this drug in research with cockatiels and case report in CA condors!

Proventriculus & Ventriculus

- Neoplasia
  - Adenocarcinoma at isthmus between proventriculus and ventriculus
  - Usually aggressive
  - Biopsy
    - Endoscopic?
    - Surgical biopsy safer?

Intestines

- Infectious
  - Bacterial – Clostridium sp., Mycobacteria sp., Chlamydia sp., Gram negative bacteria
  - Fungal – *Candida* spp
  - Parasitic
  - Viral –Psittacid herpesvirus
    - Internal papillomas, neoplasia
      - Adenocarcinoma-bile duct and pancreatic correlated with Psittacid herpesvirus?
    - Paramyxoviruses
- Strictures
- Foreign bodies

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Majority of intestinal pathology is due to infectious causes.

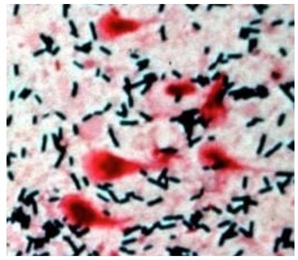
Neoplasia of the intestines may include adenocarcinoma, which may also be found in the proventriculus & isthmus.

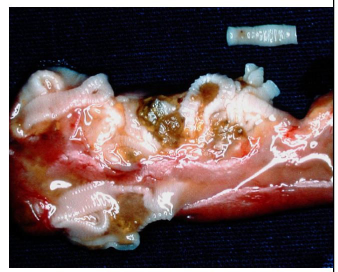
Biliary duct and pancreatic carcinomas are reported and there may be a higher incidence in birds with psittacid herpesvirus. Species prone to Psittacid herpesvirus include New World species (Macaws, Amazon parrots, conures) most commonly.

Several paramyxovirus strains – affect different species of birds. Parrots can be affected by PMV-1 and PMV-3

Intestines

- Parasitic
  - Ascarids
  - Cestodes
  - Protozoa
    - Giardia
    - Trichomonas
    - Coccidia
      - Atoxoplasma
      - Cryptosporidia
  - Hexamita, Histomoniasis





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- Ascarid parasites are a problem for birds kept on natural substrates. They can cause enteritis or GI obstructions. Cestodes were a problem in imported birds, but are rarely seen anymore. Birds kept outdoors though may have them.
- Giardia was more commonly reported in cockatiels. Clinical signs may include weight loss, failure to thrive, diarrhea, or possibly feather destructive behavior. Disease may also be inapparent. Treatment is commonly with metronidazole.

- Coccidia ranges from inapparent infection to hemorrhagic diarrhea and death. Species include Eimeria in pigeons and galliformes and Isospora in psittacines and passerines. There is a direct life cycle with transmission via ingestion of sporulated oocysts in contaminated food or water. Disease is precipitated by stress. Diagnosis is observation of oocysts on fecal float or wet mount. Treatment: coccidiostats such as ponazuril. Atoxoplasma affects the liver of passerines, and will be discussed in the hepatic lectures.

- Keeping backyard chickens with turkeys can be a problem due to Histomonas meleagridis. This protozoa can infect all poultry, but it is most deadly to turkeys. They are transported on the cecal nematode Hexamita gallinarum which can be carried by chickens, without clinical signs. So- what are the clinical findings in turkeys? This will without a doubt be on your national boards!

Cloaca

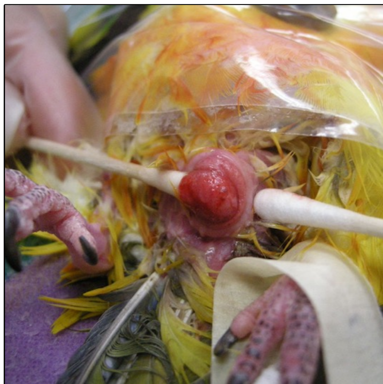
- Infectious
  - Bacterial
  - Viral
- “*Iltovirus psittacidalpha* 1”
  - Psittacid herpesvirus 1
- Neoplasia
  - Adenocarcinoma, carcinoma
- Prolapse

Infectious disease of the cloaca may be bacterial, including cloacitis due to trauma (cloacoliths or chronic prolapse), coexisting disease, or nutritional deficiencies. Image from scienceblogs.com

Cloaca

- Psittacid Herpesvirus 1
  - Pacheco's Disease of South American parrots
  - "Mucosal papillomatosis", and associated neoplasias
  - Pacheco's rarely seen since Wild Bird Conservation Act enacted to stop importation of wild parrots
    - Thoughts we had eradicated the disease by the early 2000s

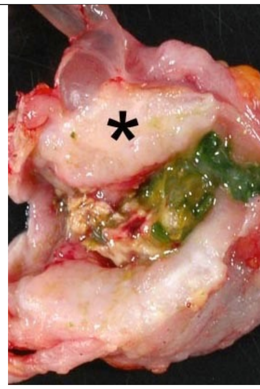
- Psittacid herpesvirus (PsHV) 1 affect primarily South American (New world) parrots.
- Pacheco's disease was first identified in the 1920s in Brazil and in the 1970s when large numbers of birds were being imported into the pet trade in the US and Europe. Rare outbreaks now, often when
  birds are mixed in an aviary.
- Pacheco's disease was peracute- birds are often dead before clinical signs are seen. Birds die from acute hepatic necrosis with this herpesvirus genotype. On necropsy livers are very enlarged and histopathology shows profound necrosis.
- This was a disease of imported birds- once the wild bird conservation act of 1992 was enacted, we saw a dwindling of cases over the years and by the early 2000s we were not seeing many cases anymore, so we patted our selves on the backs and said "look at us, eradicating a disease!"



Cloaca

- Psittacid Herpesvirus 1
  - “Mucosal Papillomatosis”
  - Affects O/P, crop, cloaca primarily
  - Associated with carcinomas of intestines, liver, pancreas?
- Tenesmus, bloody droppings (melena if proximal GIT), malodorous feces, ureteral obstruction life threatening
- Visualize lesions (may require endoscopy)
- Affected tissue blanches with acetic acid?
- Herpes PCR may or may not be positive
- Surgical removal of hemorrhaging or obstructive lesions necessary before ureters are obstructed

- Birds affected with “papillomatosis” may have cloacal mucosa that appears irregular, cobblestone appearance or there may be focal masses. This is particularly common in South American species of birds and can lead to proctodeal and urodeal obstructions.
- Clinical signs include tenesmus, bloody droppings (melena if lesions in more proximal GIT), malodorous feces, staining of vent and tail feathers with droppings.
- Most life threatening clinical finding is ureteral obstruction but in females oviductal obstruction can require emergency surgery if they are eggbound. Thus it is most important to thoroughly visualize the lesion using cloacoscopy, or at surgery.
- Internal lesions may also be present and endoscopy, contrast radiographic studies, or ultrasound may be indicated for identification of internal lesions.
- There appears to be an increased incidence of biliary, pancreatic and gastrointestinal adenocarcinomas in birds with PsHV lesions. However, a very recent paper did not find an association between PsHV1.
- Surgical therapy using laser, cryosurgery and/or radiosurgery can reduce the size of lesions, but often lesions may return. Recurrence following removal is common. There is not a single surgical treatment that is superior to another for this condition and the prognosis for long-term survival is guarded depending on the size of the mass and its location.



Cloaca

- Neoplasia
  - SCC, Carcinoma
  - Adenocarcinoma most common

- Cloacal SCCs, carcinomas, adenocarcinomas are reported and have been thought to potentially represent a malignant transformation of a papilloma. However a recent paper could not correlate papillomas with SCCs
- These tumors are locally aggressive and may also be associated with an increased incidence of bile duct or hepatic carcinomas.
- The prognosis is poor and birds may die secondary to ureter obstruction as in the Amazon parrot above.

Cloaca

- Cloacal Prolapse
  - May be secondary to GI, urinary or reproductive tract disease
    - Foreign body, urolith, egg binding
  - Behavioral
    - More prevalent in Cockatoos

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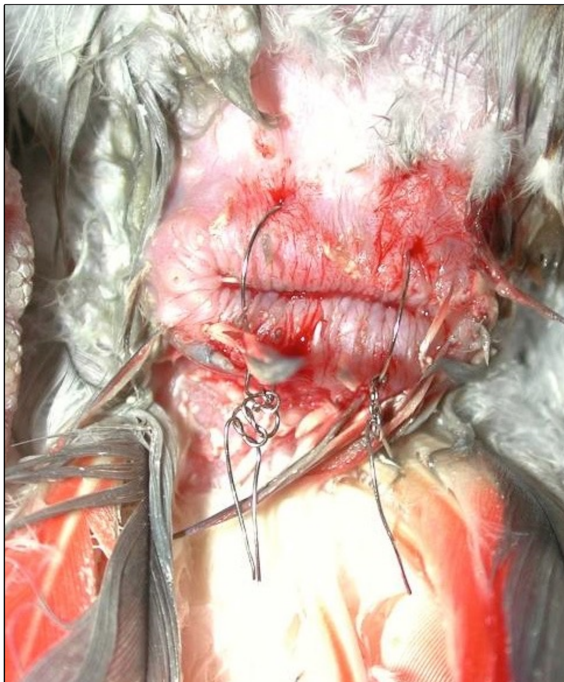
Cloacal prolapse may be seen secondary to GI or reproductive disease. We see this commonly with egg binding, foreign bodies, uroliths and cloacoliths.
It is common in cockatoos. Some birds may interpret owner's behaviors of petting them all the way down their backs as sexually stimulating. Owners often need to be educated as to the appropriate petting of their birds as this is a chronic condition and can be fatal.



Cloaca

- Cloacal Prolapse
  - Treatment
    - Analgesia first!
      - Local lidocaine jelly
    - Lavage, replace prolapsed tissue
    - Stay sutures
      - NOT purse string! WHY?
    - Repeated prolapse may require ventplasty, cloacopexy
    - Address underlying cause

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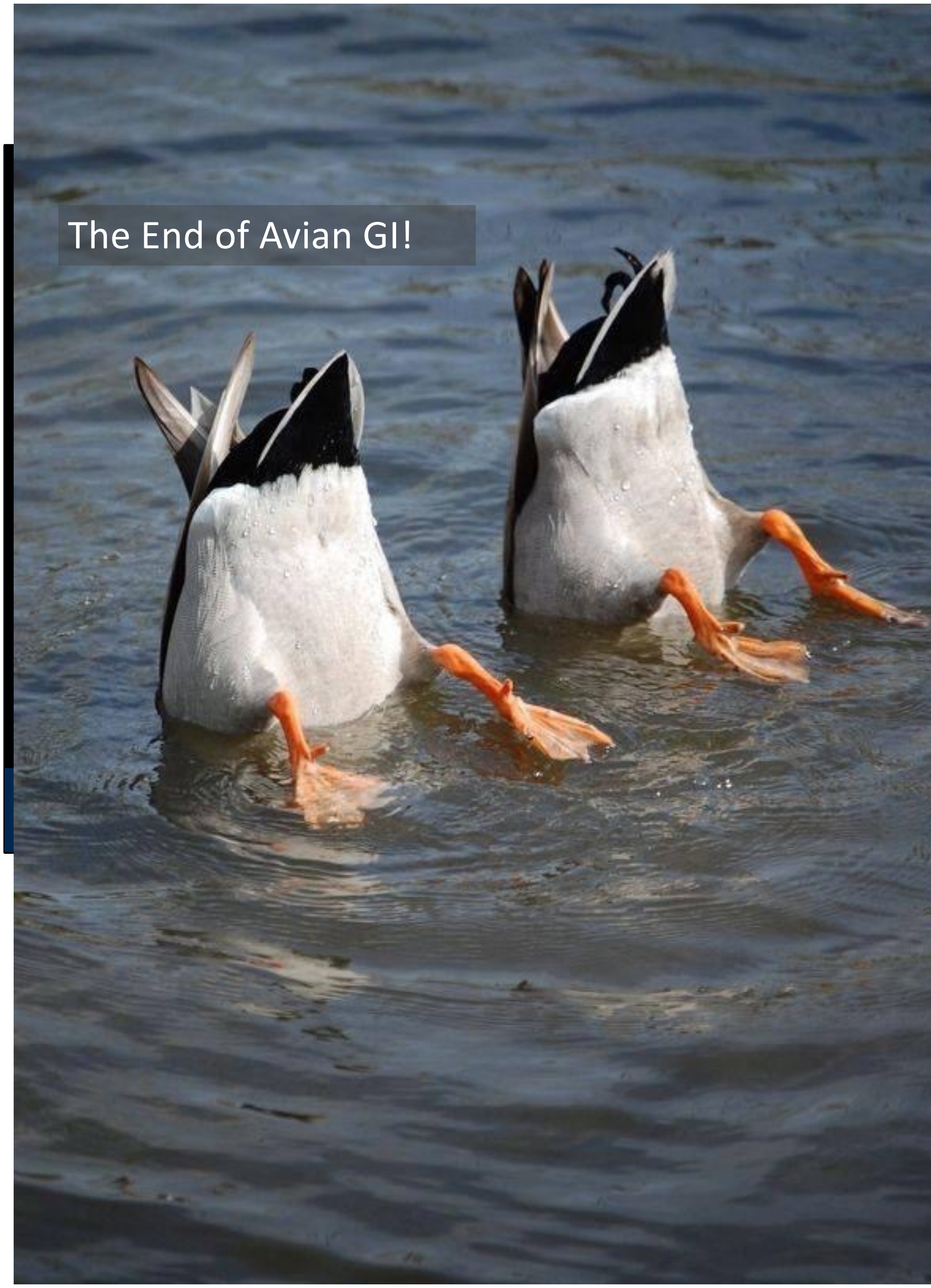
- Treatment involves positive identification of the prolapsed tissue, sometimes the oviduct will prolapse through the cloaca.
- Lavage and gently replace the tissue.
- Stay sutures may be placed on either side of the cloaca. Birds that are severely straining may require stainless steel sutures to retain the prolapsed tissue.
  - Purse string sutures are not indicated as they can permanently damage the cloacal sphincter.
- It is critical to determine the underlying cause. Birds with chronic prolapse may require surgery to reduce the vent opening (ventplasty) or cloacopexy to attach the cloaca to the internal coelomic body wall.
  - Birds will prolapse in spite of these procedures if the cause is not addressed.
  - The long-term prognosis for these birds is guarded to poor if behavioral as it may be impossible to stop if chronic.


The End of Avian GI!